5G RAN

Fusion Cell (Low-Frequency TDD) Feature Parameter Description

Issue Draft A

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-08-30).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between "time-domain power saving BWP" and "Fusion Cell". For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between "PUSCH coordinated power control" and "Fusion Cell". For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationships of "Fusion Cell" with "uplink 4-layer transmission for a single FWA user", "FWA device-pipe-based uplink dense codebooks", and "FWA device-pipe-based uplink high power". For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Updated the impact relationship between "Extended Cell Range" and "Fusion Cell". For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-091101	Virtual 128T	4 Fusion Cell

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Fusion Cell	None	**4 Fusion Cell**

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

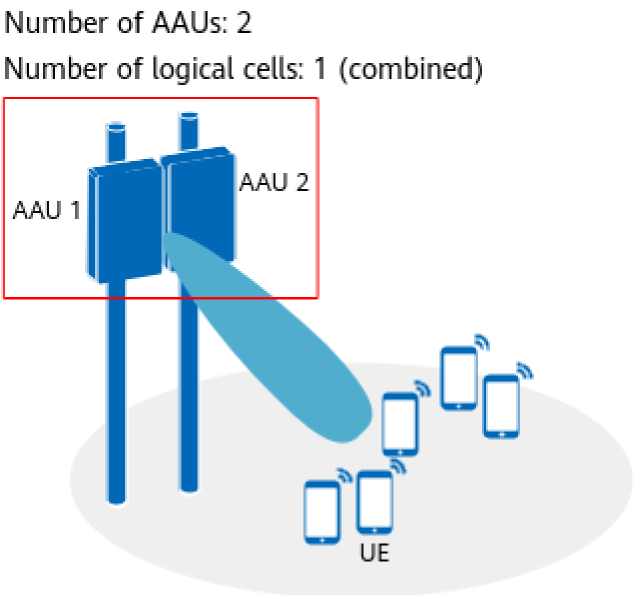
Function Name	Difference	Chapter/Section
Fusion Cell	Supported only in low frequency bands	**4 Fusion Cell**

3 Overview

To keep up with rising 5G requirements, operators need to add NR carriers for bandwidth expansion or deploy new sites for improved coverage, ensuring improved user experience. However, the limited power of existing AAUs leads to coverage shrinkage when adding NR carriers. Additionally, site acquisition in certain areas, like residential areas, is challenging, making it difficult to deploy new sites. Fusion Cell can resolve coverage shrinkage caused by limited power after carrier addition. It can also provide coverage together with base stations surrounding residential areas to tackle site acquisition challenges.

Fusion Cell combines two TRPs working on the same frequency and bandwidth on two massive MIMO AAUs into one logical cell to provide radio communication services for UEs, thereby achieving power gains and array gains of multiple antennas, as shown in **Figure 3-1**.

Figure 3-1 Fusion Cell



For details about TRPs, see *Cell Management*.

4 Fusion Cell

4.1 Principles

With Fusion Cell, inter-TRP coordinated scheduling enables the gNodeB to combine two TRPs into a logical cell to provide radio communication services for UEs. This function requires that the **NRDUCell**.***NrDuCellNetworkingMode*** parameter be set to **FUSION_CELL**.

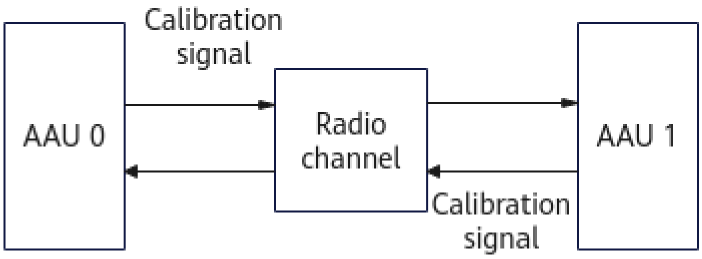
- **Over-the-Air Channel Calibration**: A cell enabled with Fusion Cell is a logical cell formed by two TRPs, and the phases of signals transmitted by the two AAUs are different. As such, joint channel calibration needs to be performed on the two TRPs. In this way, uplink and downlink reciprocity is ensured between TX and RX channels of the two TRPs, and joint beamforming can be performed to improve user experience.
- **Fusion Cell Cluster**: To specify the scope of joint channel calibration, the two TRPs of a cell enabled with Fusion Cell must be configured as a cluster. Joint channel calibration is performed on a per cluster basis.
- **Inter-TRP Coordinated Scheduling**: The gNodeB provides radio communication services for UEs through inter-TRP coordinated scheduling.

Over-the-Air Channel Calibration

Fusion Cell requires two combined AAUs. The phases of the signals transmitted by the two AAUs are different. Inter-AAU channel calibration ensures that the RF channels of the two AAUs in Fusion Cell have the same delay so that the signals transmitted over those channels of the two AAUs can be combined in the same phase on the UE side. With inter-AAU channel calibration, the base station can adjust the TX/RX response ratio of each RF channel of the AAUs to be consistent with that of the reference AAU. This ensures uplink and downlink reciprocity between TX and RX channels of two TRPs.

Figure 4-1 illustrates the principles for over-the-air channel calibration.

Figure 4-1 Over-the-air channel calibration



For details about how to check whether inter-AAU over-the-air channel calibration is successful, see **4.4.2 Activation Verification**.

Fusion Cell Cluster

The specific configurations of a Fusion Cell cluster are as follows:

- Run the **ADD NRDUCELLTRP** command to add two TRPs to a cell enabled with Fusion Cell. This is because the cell enabled with Fusion Cell must be served by two TRPs and the two TRPs must work in 64T64R mode.
- Run the **ADD GNBCLUSTER** command to add a cluster.
- Run the **ADD GNBMIMOCLUSTERCELL** command to add cells to this cluster.

  The **GNBMIMOCLUSTERCELL**.***Mcc*** and **GNBMIMOCLUSTERCELL**.***Mnc*** parameters must be set to the mobile country code (MCC) and mobile network code (MNC) of the primary operator, respectively.

Inter-TRP Coordinated Scheduling

The two TRPs for a cell enabled with Fusion Cell share air interface resources as follows:

- The PUSCH and PDSCH are scheduled in a unified manner, and the total resources are the same as those in a common cell.
- For the uplink common channels and signals, such as the PUCCH, PRACH, and sounding reference signal (SRS), resources are allocated in a unified manner in the cell.
- The **NRDUCellCsirs**.***InterTrpCsirsStaggerMode*** parameter must be set to **NOT_STAGGER** for CSI-RS resources in a cell enabled with Fusion Cell. This is because tracking reference signal (TRS) and CSI-RS resources are allocated in a unified manner in the cell.
- For the PBCH and synchronization signal (SS), the base station uses the same time-domain positions for SSB transmissions from two TRPs to improve the SSB coverage of the cell.

4.1.1 Fusion Cell Enhancements

User experience in a cell enabled with Fusion Cell is improved by increasing the spectral efficiency and channel calibration success rate and adjusting the SSB coverage range of the cell.

MU-MIMO UE SINR Enhancement

In low-frequency TDD, two TRPs share air interface resources in a cell enabled with Fusion Cell, and resources are allocated in a unified manner in the cell. However, RB resources are still limited in medium- and heavy-load scenarios, affecting user experience. To improve the downlink user experience when resources are limited, MU-MIMO UE SINR enhancement is introduced. This function is controlled by the **MUMIMO_SINR_ENH_SW** option of the **NRDUCellMultiTrp**.***FusionCellAlgoSwitch*** parameter. When this function is enabled, the SINR calculation method for MU-MIMO UEs is modified in a cell enabled with Fusion Cell. The signal modulation order is increased based on the power gains brought by coordinated scheduling between two TRPs, thereby improving the frequency usage and downlink system capacity.

Interference Avoidance for Inter-TRP Calibration in a Cell Enabled with Fusion Cell

To better avoid interference for a cell enabled with Fusion Cell and increase the inter-TRP calibration success rate by adjusting time-frequency signal resources during inter-AAU channel calibration, "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell" is introduced. This function is controlled by the **FUSION_CALIB_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***ChnCalibPolSw*** parameter.

When this function is enabled in the networking of multiple cells enabled with Fusion Cell, the gNodeB flexibly adjusts time-frequency signal resources among these cells to avoid calibration interference between Fusion Cell clusters, as shown in **Figure 4-2**.

When cells enabled with Fusion Cell and common cells enabled with Co-MM are configured alternately, "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell" can also avoid calibration interference between Fusion Cell clusters and Co-MM clusters, as shown in **Figure 4-3**. Co-MM is controlled by the **INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter. For details about Co-MM, see *Co-MM (Low-Frequency TDD)*.

Figure 4-2 Principle of "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell"

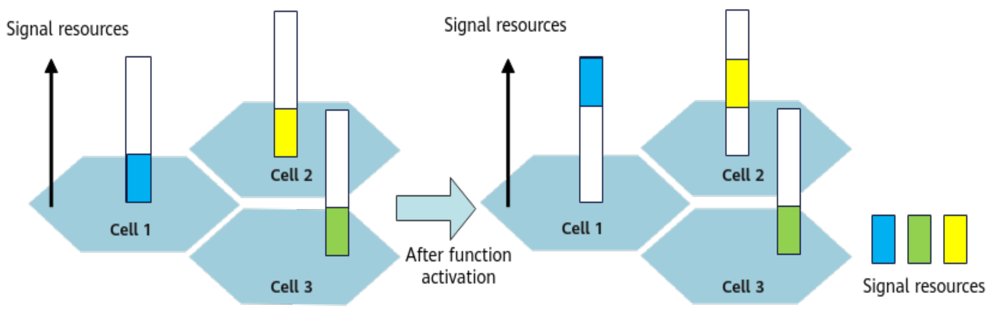
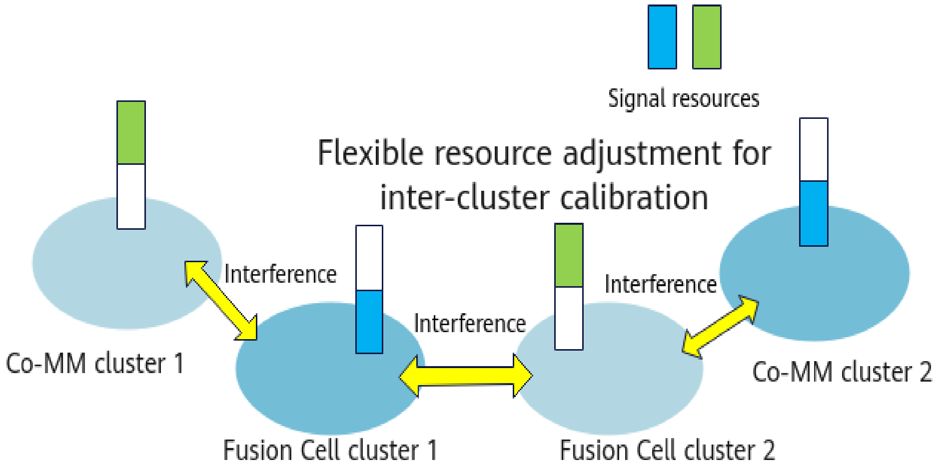


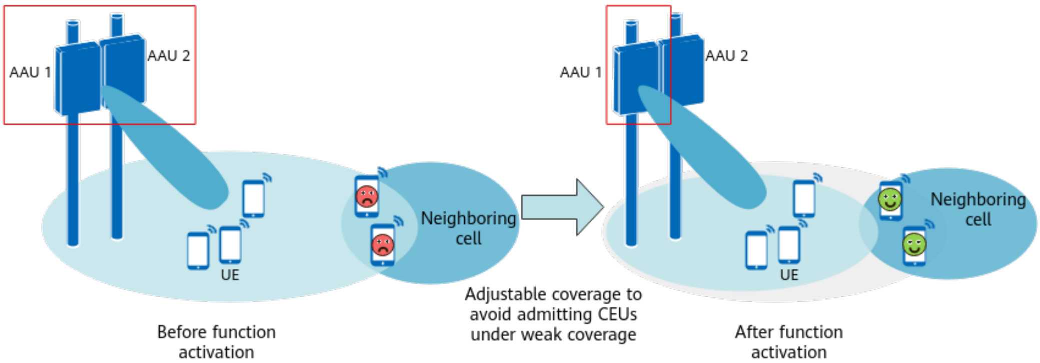
Figure 4-3 Interference avoidance for calibration between Fusion Cell clusters and Co-MM clusters



Single-TRP SSB Transmission

In low-frequency TDD, a cell enabled with Fusion Cell combines two TRPs to form a logical cell using inter-TRP coordinated scheduling. The two TRPs jointly transmit SSBs at the same time-domain position to improve the cell SSB coverage. Single-TRP SSB transmission is introduced to implement coverage adjustment. This function prevents a cell enabled with Fusion Cell from admitting some CEUs under weak coverage, thereby preventing user experience deterioration. This function is controlled by the **SINGLE_TRP_SSB_TRANS_SW** option of the **NRDUCellMultiTrp**.***FusionCellAlgoSwitch*** parameter. When this function is enabled, a single TRP is used for SSB transmission in a cell enabled with Fusion Cell to narrow down the coverage area, as shown in **Figure 4-4**.

Figure 4-4 Principle of single-TRP SSB transmission in a cell enabled with Fusion Cell



4.2 Network Analysis

4.2.1 Benefits

Compared with common cells, cells enabled with Fusion Cell experience improvements in cell coverage and uplink and downlink UE throughputs.

Compared with cells enabled with Fusion Cell, cells enabled with Fusion Cell enhancements experience the following benefits:

- After MU-MIMO UE SINR enhancement is enabled, there are increases in the average downlink UE throughput (**User Downlink Average Throughput (DU)**) and the proportion of PRBs paired for downlink MU-MIMO (**N.ChMeas.MIMO.DL.Pair.PRB**/(**N.PRB.DL.Used.Avg** x **N.DL.PDSCH.Tti.Num**) x 100%) if the following conditions apply in medium- and heavy-load scenarios (50% ≤ Downlink PRB usage (**N.PRB.DL.Used.Avg**/ **N.PRB.DL.Avail.Avg**) ≤ 95%): (1) The cell resources are limited; (2) the proportion of PRBs paired for downlink MU-MIMO is greater than 10%. There are fewer or no benefits in the following scenarios:
  - Light-load scenario (with the downlink PRB usage being no greater than 50%)
  - Scenario where the majority of UEs are located at the cell center or at a medium distance from the cell center (for example, with the average CQI of a cell being greater than 12)
  - Scenario with a single traffic model for UEs (for example, with the proportion of tail packet transmission duration being less than 20% or greater than 85%)
- In the networking where multiple cells enabled with Fusion Cell are deployed, enabling "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell" increases the inter-TRP calibration success rate. When cells enabled with Fusion Cell and common cells enabled with Co-MM are configured alternately, enabling "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell" can further increase the inter-TRP calibration success rate.
- After single-TRP SSB transmission is enabled and the coverage of a cell remains the same before and after Fusion Cell is enabled, the cell does not admit some CEUs under weak coverage. This prevents the CQI and some accessibility-related indicators from deteriorating.

  NOTE

  - Any two different AAU models can be combined. However, due to differences in different models, it is recommended that the same module be used for combination. Otherwise, the performance after combination may be affected and the expected gains cannot be achieved.
  - It is recommended that the maximum transmit power, maximum SSB power offset, TRS power offset, and maximum value of the aggregated power offset on the PDSCH be consistent between the two combined AAUs. Otherwise, the expected gains of Fusion Cell may be affected. The specific parameters are as follows:
    - **NRDuCellTrp**.***MaxTransmitPower***
    - **NRDUCellTrpBeam**.***MaxSsbPwrOffset***
    - **NRDUCellChnPwr**.***TrsPwrOffset***
    - **NRDUCellChnPwr**.***MaxPdschConvPwrOffset***

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

Compared with common cells, cells enabled with Fusion Cell experience the following impacts:

- The coverage is improved, and the cell admits some CEUs under weak coverage. As a result, the CQI value or the values of certain accessibility-related indicators may decrease, and the uplink and downlink CCE aggregation levels and the uplink and downlink CCE allocation failure rates increase.
- The values of counters related to uplink cell interference measurement, such as **N.UL.NI.Avg.PRB0** to **N.UL.NI.Avg.PRB272** and **N.UL.NI.Max.PRB0** to **N.UL.NI.Max.PRB272**, may be greater than expected.
- The values of counters related to channel quality measurement, such as **N.UL.PUSCH.RSRP.Index0** to **N.UL.PUSCH.RSRP.Index11**, **N.UL.PUSCH.RSSI.Index0** to **N.UL.PUSCH.RSSI.Index4**, and **N.UL.PUSCH.SINR.Index0** to **N.UL.PUSCH.SINR.Index7**, will double.
- The values of counters related to NR inter-carrier dynamic power sharing, such as **N.PwrShare.PwrIn.Slot.Num** and **N.PwrShare.PwrOut.Slot.Num**, will double.
- The value of the **N.PowerSaving.SymbolShutdown.Dur** counter will double.
- The scheduling capability of cells enabled with Fusion Cell is enhanced and the PRB usage decreases. As a result, the proportion of PRBs paired for downlink MU-MIMO may decrease.

Compared with cells enabled with Fusion Cell, cells enabled with Fusion Cell enhancements experience the following impacts:

- After single-TRP SSB transmission is enabled, the coverage decreases but the coverage of a cell remains the same before and after Fusion Cell is enabled. The number of UEs in a cell enabled with Fusion Cell decreases, which can be observed using the **Average User Number (CU)** and **Maximum User Number(CU)**.
- MU-MIMO UE SINR enhancement has other possible network impacts as follows:
  – After an increase in the proportion of PRBs paired for downlink MU-MIMO, there may be a decrease in the average downlink UE throughput if the data volume of tail packets accounts for a large proportion in medium-load scenarios or if the channel quality of UEs is good and UEs are distributed in a centralized manner.
  – After an increase in the proportion of PRBs paired for downlink MU-MIMO, there will be decreases in the average downlink MCS index and average rank of UEs, and a possible decrease in the downlink throughput of some UEs.
  – After an increase in the proportion of PRBs paired for downlink MU-MIMO, there will be an increase in the interference between paired UEs and accordingly an increase in the downlink BLER in the cell.
  – After an increase in the proportion of PRBs paired for downlink MU-MIMO, there will be increases in the number of scheduled UEs in the downlink and the consumption of CCE resources. If CCE resources for the uplink are limited, there will be an increase in the CCE allocation failure rate for uplink transmission and decreases in the average uplink cell throughput and average uplink UE throughput.
  – After an increase in the proportion of PRBs paired for downlink MU-MIMO, there will be an increase in the number of uplink status reports. If

PUSCH PRB resources are limited, there will be decreases in the average uplink cell throughput and average uplink UE throughput.

– The PRB utilization may decrease in medium- and heavy-load scenarios. It fluctuates normally in light-load scenarios.

4.3 Requirements

4.3.1 Licenses

None

NOTE

- For other features whose sales units are "per Cell" for the TRPs serving a cell enabled with Fusion Cell, the license consumption is based on the number of TRPs. For example, if a cell enabled with Fusion Cell contains two TRPs, two licenses are consumed.
- If a TRP needs to be added, check whether a capacity license is required. For details about the capacity licenses, see *License Management*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- Fusion Cell

Function Name	Function Switch	Reference	Description
UE-characteristic-based SRS period adaptation	**USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.*SrsAlgoSwitch*** parameter	*Channel Management*	Fusion Cell requires that UE-characteristic-based SRS period adaptation be enabled.
Inter-TRP CSI-RS Stagger Mode	**NRDUCellCsirs.*InterTrpCsirsStaggerMode***	None	This parameter must be set to **NOT_STAGGER**.

Function Name	Function Switch	Reference	Description
Downlink Maximum MIMO Layer Number Quota	**NRDUCellPdsch**.***MaxMimoLayerNum***	*MIMO (TDD)*	This parameter must be set to **LAYER_2**, **LAYER_4**, **LAYER_6**, **LAYER_8**, **LAYER_10**, **LAYER_12**, **LAYER_14**, **LAYER_16**, or **LAYER_DEFAULT**.
Slot Assignment	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	The **NRDUCell**.***SlotAssignment*** parameter must be set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**. The number of GP symbols in a self-contained slot must be 17, 18, or within the range of 2 to 6. For details about the setting of the number of GP symbols in self-contained slots with different slot configurations, see *Standards Compliance*.

- Fusion Cell enhancements

Function Name	Function Switch	Reference	Description
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Fusion Cell is required for enabling MU-MIMO UE SINR enhancement and single-TRP SSB transmission.
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Fusion Cell is required for "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell" to take effect.

4.3.2.2 Mutually Exclusive Functions

- Fusion Cell

  Some features or functions may not be supported when certain NR DU cell networking modes are used. For details about the cell requirements of these features or functions, see section "Requirements" in the corresponding feature documents.

Function Name	Function Switch	Reference
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*
Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** or **INTER_GNB_DL_CS_CBF_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*
Downlink hybrid precoding	**DL_HYBRID_PRECODING_SW** option of the **NRDUCellPdschPrecode**.***PrecodingAlgoSwitch*** parameter	*MIMO (TDD)*
PUSCH coordinated power control	**PUSCH_COORD_PWR_CTRL_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*
Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch.*MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*
Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*

Function Name	Function Switch	Reference
SCC coverage expansion	**SA_SCC_COV_THLD_ADAPT_SW** and **NSA_SCC_COV_THLD_ADAPT_SW** options of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*
Single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Aggregation*
CA Performance Boost Based on Multi-dimensional Optimization	**CA_PRFM_BOOST_BSD_ON_MD_OPT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*Multi-Frequency Smart Aggregation*
Differentiated eMBB QoS	**EMBB_QOS_DIFF_SW** option of the **NRDUCELLALGOSWITCH**.***Lowlatency Switch*** parameter	None
Interference coordination between public and dedicated networks	**HIGH_SPEED_INTRF_AVOID_SW** option of the **NRCellAlgoSwitch.*HighSpeedDedPolicySw*** parameter	*High Speed Mobility*
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*
LTE TDD and NR Flash Dynamic Power Sharing	**LTE_NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*LTE and NR Power Allocation*
NR multi-band power adaptive scheduling	**NR_MULTI_BAND_PWR_ADAPT_SCH_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*NR Inter-Carrier Dynamic Power Allocation*
Independent SUL deployment	**SUPER_UL_TDM_SCH_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*
Slot-level power control function for smart power lock	**POWER_CONTROL_SLOT_SW** option of the **NRDUCellSmartPwr-Lock**.***SmartPowerLockSw*** parameter	None

Function Name	Function Switch	Reference
LTE and NR joint power management	**gNBDUSpsGrp**.*SpsType* set to **LNR_INTER_RAT_SPS**	None
Intra-TTI time-frequency-power collaboration	**INTRA_TTI_TFP_COLLAB_SW** option of the **NRDUCellAlgoSwitch**.*PowerSavingSwitch* parameter	None
UL and DL Decoupling	**NRDUCellAlgoSwitch**.*UlDlDecouplingSwitch*	*UL and DL Decoupling*
R15 subband type II PMI	**R15_TYPE2_SW** option of the **NRDUCellPdschPrecode**.*DlCodebookTypeSwitch* parameter	None
R16 subband type II PMI	**R16_TYPE2_SW** option of the **NRDUCellPdschPrecode**.*DlCodebookTypeSwitch* parameter	None
Baseband resource mutual aid	**NRDUCellTrp**.*BbResMutualAidSw* set to **ON**	None
Uplink RF combining noise suppression	**RF_COMBINING_NOISE_SUPPR_SW** option of the **NRDUCellUlSch**.*RfCombSwitch* parameter	None
Inter-TTI time-frequency-power collaboration	**INTER_TTI_TFP_COLLAB_SW** option of the **NRDUCellAlgoSwitch**.*PowerSavingExtSwitch* parameter	None
Intra-frequency beam statistics	**INTRA_FREQ_BEAM_STAT_SW** option of the **NRDUCellCollabServ**.*NeighborNrCellMgmtSw* parameter	None
Closed-loop PUSCH antenna selection	**PUSCH_CLOSED_LOOP_ANT_SEL_SW** option of the **NRDUCellChnCovAlgo**.*UlCoverageAlgoSwitch* parameter	*Turbo Uplink (Low-Frequency TDD)*
Format-1 HARQ-ACK resource scheduling in RB saving mode	**HARQ_ACK_RB_SAVING_SCH_SW** option of the **NRDUCellPucch**.*SrResoureAlgoSwitch* parameter	*Channel Management*
3D coverage pattern[a]	**NRDUCellTrpBeam**.*CoverageScenario*	*Beam Management*

Function Name	Function Switch	Reference
Downlink single-UE robust mobility precoding algorithm	**DL_SU_ROBUST_MOB_PRECODE_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	None
Interference-based subband robust weights	**INTRF_BASED_ROBUST_WEIGHT_SW** option of the **NRDUCellIntrfIdent**.***ChnMeasCtrlSw*** parameter	*Interference Avoidance*
SRS antenna selection for RedCap users in FWA scenarios	**REDCAP_SRS_ANT_SWITCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*FWA*
Downlink 8-layer transmission for a single FWA user	**DL_SU_EIGHT_LAYER_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter	*FWA*
Uplink 4-layer transmission for a single FWA user	**UL_SU_FOUR_LAYER_SW** option of the **NRDUCellUlRank**.***UlRankAlgoSw*** parameter	*FWA*
FWA device-pipe-based uplink dense codebooks	**UL_DENSER_CODEBOOK_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter	*FWA*
FWA device-pipe-based uplink high power	**UL_HIGH_POWER_UE_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter	*FWA*
Time-domain interference avoidance	**SS54_1GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter	*Interference Avoidance*
Uplink symbol power saving	**UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*
Dynamic RF channel shutdown	**RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*

Function Name	Function Switch	Reference
Optimal channel shutdown	**TTI_OPTIMAL_RF_CHN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*
Flexible spectrum access	**FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt**.***MultiBandFlexSchSwitch*** parameter	*Uplink Flexible Spectrum Scheduling*
Note: a: If the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **FUSION_CELL** for a cell, the **NRDUCellTrpBeam**.***CoverageScenario*** parameter corresponding to the cell can only be set to **DEFAULT**, **SCENARIO_1** to **SCENARIO_16**, **EXPAND_SCENARIO_1**, **CUSTOMIZED_BEAM_SCENARIO**, or a value indicating 3D coverage scenarios for emergency base stations (**SCENARIO_33**, **SCENARIO_34**, or **SCENARIO_35**). For descriptions of emergency base stations, see *Emergency Communications Solution (Low-Frequency TDD)*.		

– Fusion Cell enhancements

None

4.3.2.3 Function Impacts

- Fusion Cell

Function Name	Function Switch	Reference	Description
NR inter-carrier dynamic power sharing	**NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*NR Inter-Carrier Dynamic Power Allocation*	In networking enabled with Fusion Cell, the number of scenarios where NR inter-carrier dynamic power sharing takes effect may decrease.

Function Name	Function Switch	Reference	Description
Downlink power control	**NRDuCellTrp**.***MaxTransmitPower*** **NRDUCellTrpBeam**.***MaxSsbPwrOffset*** **NRDUCellChnPwr**.***TrsPwrOffset*** **NRDUCellChnPwr**.***MaxPdschConvPwrOffset***	*Power Control*	It is recommended that the maximum transmit power, maximum SSB power offset, TRS power offset, and maximum value of the aggregated power offset on the PDSCH be consistent between the two combined AAUs. Otherwise, the gains of Fusion Cell may be affected.
RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	If the **NRDUCellTrpBeam**.***CoverageScenario*** parameter is set to **EXPAND_SCENARIO_1** for a cell enabled with Fusion Cell, SSBs cannot obtain power gains brought by Fusion Cell after RF channel intelligent shutdown takes effect.
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	When there is a large proportion of UEs in power saving BWP, the gains on the downlink user-perceived rate brought by Fusion Cell are affected.
Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	When there is a large proportion of BWP2 UEs, the gains on the downlink user-perceived rate brought by Fusion Cell are affected.
Downlink initial beam selection	**DL_INITIAL_BEAM_SELECT_SW** option of the **NRDUCellAlgoSwitch**.***BeamOptSwitch*** parameter	*Beam Management*	It is recommended that this option be selected for cells enabled with Fusion Cell. Otherwise, the access success rate may decrease.

Function Name	Function Switch	Reference	Description
3D coverage pattern	**NRDUCellTrpBeam**.***CoverageScenario***	*Beam Management*	It is recommended that the settings of the following parameters be consistent between the two TRPs serving a cell enabled with Fusion Cell to obtain complete coverage gains: ● **NRDUCellTrpBeam**.***CoverageScenario*** ● **SCENARIO_BEAM_DENSIFY_ALGO_SW** option of the **NRDUCellTrpBeam**.***ScenarioBeamAlgoSw*** parameter ● **NRDUCellTrpBeam**.***Azimuth*** and **NRDUCellTrpBeam**.***Tilt***
Extended Cell Range	**NRDUCell**.***CellRadius*** set to a value greater than **14500**	*Extended Cell Range*	If Fusion Cell is required to be enabled in a cell with Extended Cell Range enabled and the **NRDUCell**.***SlotStructure*** parameter set to **SS518** or **SS818**, then the **SS518_4GP_TIME_INTRF_AVOID_SW** or **SS818_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter corresponding to the cell must be selected.
Simulated load	**STR NRDUCELLSIMULATE-LOAD**	None	Simulated load is not supported in cells enabled with Fusion Cell.

- Fusion Cell enhancements

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards can be used.

NR TDD-capable baseband processing units: UBBPg2a, UBBPg3, UBBPg3a, UBBPg3b, UBBPh3a, UBBPh3b, and UBBPh3

The configuration requirements on the baseband processing unit are described as follows:

- The UBBPg3 and UBBPg3a support both LTE and NR. Other UBBPg series baseband processing units support only NR and can serve only TDD cells.
- UBBPh series baseband processing units support both LTE and NR and serve only TDD cells.
- The two TRPs serving a cell enabled with Fusion Cell are configured on the same baseband processing unit.
- The maximum number of cells enabled with Fusion Cell supported by the baseband processing unit is half of the maximum number of NR 64T common cells. (If there is a decimal, round it down.)
- Cells enabled with Fusion Cell and common cells can be configured on the same baseband processing unit, but cells enabled with Fusion Cell and SUL cells cannot be configured on the same baseband processing unit.

RF Modules

The RF modules must meet all the following requirements:

- The TX/RX mode is 64T64R.
- The eCPRI protocol is supported.
- NR TDD is supported.
- RF modules work in the n38, n41, n77, or n78 frequency band.
- MetaAAUs are not used.

NOTE

For details about RF module requirements, see "Technical Specifications" and "Hardware Description" of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Networking

None

4.3.5 Cells

None

4.3.6 Others

- It is recommended that two AAUs be deployed in the same direction and on the same plane, and the angle between the normal lines of the two AAUs be less than 3 degrees. They can be assembled horizontally and vertically with a spacing of less than 2 m.
- In the scenario where two AAUs of different models are combined, the beam scenario configured by the two AAUs and the tilt and azimuth corresponding to the beam scenario must be within the intersection range supported by the two AAUs. Otherwise, cells enabled with Fusion Cell may fail to be set up or performance gains may decrease.

  For example, if AAU 1 supports only the beam scenario specified by the **DEFAULT** value, and AAU 2 supports both the beam scenarios specified by the **DEFAULT** and **CUSTOMIZED_BEAM_SCENARIO** values, the beam scenarios of the two AAUs must be set to **DEFAULT** when the two AAUs are combined. Otherwise, cells enabled with Fusion Cell may fail to be set up or performance gains may decrease.

  For details about the beam scenarios supported by each module and the tilt and azimuth adjustment ranges corresponding to the beam scenarios, see *Beam Management*.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

For details about data preparation for cell setup (for example, adding sectors, sector equipment, and baseband equipment), see *Cell Management*. The following provides only the differences in data preparation.

Table 4-1 describes the parameters used for function activation.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell Networking Mode	**NRDUCell**.*NrDuCellNetworkingMode*	Set this parameter to **FUSION_CELL** when Fusion Cell is required.
Cluster ID	**gNBCluster**.*ClusterId*	Set this parameter based on the network plan.
Cluster Type	**gNBCluster**.*ClusterType*	Set this parameter to **INTRA_CELL_MIMO** when Fusion Cell is required.
Cluster ID	**gNBMimoClusterCell**.*ClusterId*	Set this parameter based on the network plan.
Mobile Country Code	**gNBMimoClusterCell**.*Mcc*	Set this parameter based on the network plan.
Mobile Network Code	**gNBMimoClusterCell**.*Mnc*	Set this parameter based on the network plan.
gNodeB ID	**gNBMimoClusterCell**.*gNodeBId*	Set this parameter based on the network plan.
Cell ID	**gNBMimoClusterCell**.*CellId*	Set this parameter based on the network plan.
Fusion Cell Algorithm Switch	**NRDUCellMultiTrp**.*FusionCellAlgoSwitch*	It is recommended that the **MUMIMO_SINR_ENH_SW** option be selected in medium- and heavy-load scenarios.
Channel Calibration Policy Switch	**gNodeBAlgo**.*ChnCalibPolSw*	It is recommended that the **FUSION_CALIB_INTRF_AVOID_SW** option be selected in the networking where multiple cells enabled with Fusion Cell are deployed.

Parameter Name	Parameter ID	Setting Notes
Fusion Cell Algorithm Switch	**NRDUCellMultiTrp**.***FusionCellAlgoSwitch***	It is recommended that the **SINGLE_TRP_SSB_TRANS_SW** option be selected when there are a large number of CEUs with weak coverage in medium- and heavy-load scenarios.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

During deployment of cells to be enabled with Fusion Cell or TRP redeployment of cells enabled with Fusion Cell, if one of the following situations occurs, modify the networking by referring to the deployment suggestions described in **4.3.3 Hardware**. If any TRP is unavailable after the modification, run the **STR NRDUCELLREDEPLOY** or **STR BRDBBRESREALLOC** command to redeploy cells on baseband processing units. Within 15 minutes after either command is executed, if the TRPs of the NR DU cell are unavailable and redeployment is required due to baseband resource conflicts, cells deployed on the corresponding baseband equipment will be automatically restarted and redeployed. In some conflict scenarios, the redeployment command may need to be executed for two or three times. After that, if any TRP is unavailable and the cell specifications of baseband processing units are sufficient, contact Huawei engineers for help.

- Situation 1: When a new cell is set up on a BBU serving a cell enabled with Fusion Cell, the new cell fails to be activated and the alarm cause is "Redeployment required due to baseband resource conflicts".
- Situation 2: A TRP for an existing cell enabled with Fusion Cell is redeployed on another baseband processing unit. Cell activation is executed when the specifications of the baseband processing unit are sufficient. However, an alarm indicating TRP unavailability is reported with the alarm cause of "Redeployment required due to baseband resource conflicts".

The **STR NRDUCELLREDEPLOY** command is used to redeploy NR DU cells on the baseband equipment in one click. This is a high-risk MML command and running the command causes some cells served by the BBU to be reestablished. For detailed operations and precautions, see the related MML command reference.

It is recommended that all cells to be served by the BBU be configured before activating any cell to avoid alarm reporting with the cause of "Redeployment required due to baseband resource conflicts".

Activation Command Examples

- Fusion Cell

```
//Adding a cell enabled with Fusion Cell
ADD NRCELL: NrCellId=90, CellName="90", CellId=90, FrequencyBand=N78, DuplexMode=CELL_TDD,
UserLabel="fusionCell";
ADD NRDUCELL: NrDuCellId=90, NrDuCellName="90", DuplexMode=CELL_TDD, CellId=90,
PhysicalCellId=90, FrequencyBand=N78, UlNarfcn=636666, DlNarfcn=636666,
UlBandwidth=CELL_BW_100M, DlBandwidth=CELL_BW_100M, SlotAssignment=8_2_DDDDDDDSUU,
SlotStructure=SS54, TrackingAreaId=0, SsbFreqPos=7880, NrDuCellNetworkingMode=FUSION_CELL,
LogicalRootSequenceIndex=0, CustomizedBwConfigInd=NOT_CONFIG,
DeployPosition=DEPLOY_INTEGRATED;

//Adding two TRPs for the cell enabled with Fusion Cell
ADD NRDUCELLTRP: NrDuCellTrpId=90, TrpType=DEFAULT, NrDuCellId=90, TxRxMode=64T64R,
BasebandEqmId=5, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=300,
CpriCompression=3DOT2_COMPRESSION;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=90, NrDuCellCoverageId=90, SectorEqmId=90,
PowerConfigMode=TRANSMIT_POWER_DBM;

ADD NRDUCELLTRP: NrDuCellTrpId=91, TrpType=DEFAULT, NrDuCellId=90, TxRxMode=64T64R,
BasebandEqmId=5, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=300,
CpriCompression=3DOT2_COMPRESSION;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=91, NrDuCellCoverageId=91, SectorEqmId=91,
PowerConfigMode=TRANSMIT_POWER_DBM;

//Adding a gNodeB cluster
ADD GNBCLUSTER: ClusterId=90, ClusterType=INTRA_CELL_MIMO;
ADD GNBMIMOCLUSTERCELL: ClusterId=90, Mcc="302", Mnc="220", gNodeBId=1, CellId=90;

//Activating the cell
ACT NRCELL: NrCellId=90;
```

- Fusion Cell enhancements

```
//Enabling MU-MIMO UE SINR enhancement
MOD NRDUCELLMULTITRP: NrDuCellId=90, FusionCellAlgoSwitch=MUMIMO_SINR_ENH_SW-1;
//Enabling "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell"
MOD GNODEBALGO: ChnCalibPolSw=FUSION_CALIB_INTRF_AVOID_SW-1;
//Enabling single-TRP SSB transmission
MOD NRDUCELLMULTITRP: NrDuCellId=90, FusionCellAlgoSwitch=SINGLE_TRP_SSB_TRANS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Fusion Cell enhancements

```
//Disabling MU-MIMO UE SINR enhancement
MOD NRDUCELLMULTITRP: NrDuCellId=90, FusionCellAlgoSwitch=MUMIMO_SINR_ENH_SW-0;
//Disabling "interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell"
MOD GNODEBALGO: ChnCalibPolSw=FUSION_CALIB_INTRF_AVOID_SW-0;
//Disabling single-TRP SSB transmission
MOD NRDUCELLMULTITRP: NrDuCellId=90, FusionCellAlgoSwitch=SINGLE_TRP_SSB_TRANS_SW-0;
```

- Fusion Cell

```
//Deactivating the cell
DEA NRCELL: NrCellId=90;

//Retaining only one TRP for a common cell and deleting the other TRPs
RMV NRDUCELLCOVERAGE: NrDuCellTrpId=91, NrDuCellCoverageId=91;
```

```
RMV NRDUCELLTRP: NrDuCellTrpId=91;

//Removing the gNodeB cluster
RMV GNBMIMOCLUSTERCELL: ClusterId=90, Mcc="302", Mnc="220", gNodeBId=1, CellId=90;
RMV GNBCLUSTER: ClusterId=90;

//Changing the cell networking mode to common cell
MOD NRDUCELL: NrDuCellId=90, NrDuCellNetworkingMode=NORMAL_CELL;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Checking Cell Status

Run the **DSP NRDUCELL** command to check the value of **NR DU Cell State**. If the output shows that the value is **Normal**, the cell has been successfully set up.

Run the **DSP NRDUCELLTRP** command to check the value of **NR DU Cell TRP State**. If the output shows that the value is **Available**, the TRP has been successfully set up.

NOTE

You can perform the following operations to verify and query whether the parameter settings are consistent between the two TRPs serving a cell enabled with Fusion Cell:

1. Run the **STR NRCELLCFGCHK** command with **NRCellCfgChk.*CheckType*** set to **FUSION_CELL_TRP** to start the consistency check for the configurations of the two TRPs serving a cell enabled with Fusion Cell.
2. Run the **DSP NRCELLCFGCHK** command with **NRCellCfgChk.*CheckType*** set to **FUSION_CELL_TRP** to query the verification result.

Over-the-Air Channel Calibration Result Query

Run the **DSP GNBMIMOCLUSTERCELL** command to check the over-the-air channel calibration result. If the value of **Cluster Cell Status** is **Normal**, the cells have entered over-the-air channel calibration. If the value of **Inter-TRP Calibration Result** is **Success**, the over-the-air channel calibration is successful in the current period; otherwise, the calibration is unsuccessful in the period.

Fusion Cell Effective Status

If the cell status is normal and over-the-air channel calibration is successful, Fusion Cell automatically takes effect.

4.4.3 Network Monitoring

You can evaluate the gains of Fusion Cell by comparing the values of the following indicators before and after this function is enabled:

- **User Downlink Average Throughput (DU)** (The value will increase.)
- **User Uplink Average Throughput (DU)** (The value will increase.)
- **Average User Number (CU)** and **Maximum User Number(CU)** (The indicators are used in scenarios where the number of UEs in a cell increases when the coverage improves.)

You can evaluate the gains of Fusion Cell enhancements by comparing the values of the following indicators before and after these functions are enabled:

- MU-MIMO UE SINR enhancement: **User Downlink Average Throughput (DU)** and downlink proportion of PRBs paired for MU-MIMO (The values will increase.)
- Interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell: Channel calibration success rate (The value may increase and it is calculated based on the value of **Inter-TRP Calibration Result** in the **DSP GNBMIMOCLUSTERCELL** command output.)
- Single-TRP SSB transmission: **Average User Number (CU)** and **Maximum User Number(CU)** (The number of UEs in a cell enabled with Fusion Cell, as indicated by the indicators, will decrease.)

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the product documentation delivered with that version.

FAQ: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Cell Management*
- *NR Inter-Carrier Dynamic Power Allocation*
- *Channel Management*
- *CoMP*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Multi-Frequency Smart Selection*
- *Multi-Frequency Smart Aggregation*
- *Cell Combination*
- *High Speed Mobility*
- *LTE and NR Power Allocation*
- *Super Uplink*
- *UL and DL Decoupling*
- *Energy Conservation and Emission Reduction*
- *Cell Combination*
- *Turbo Uplink (Low-Frequency TDD)*
- *Beam Management*
- *Power Control*
- *Co-MM (Low-Frequency TDD)*
- *Standards Compliance*
- *License Management*
- *Beam Management*
- *Extended Cell Range*
- *UE Power Saving*
- *FWA*
- *MIMO (TDD)*
- *Uplink Flexible Spectrum Scheduling*
- *Interference Avoidance*

- *Emergency Communications Solution (Low-Frequency TDD)*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*